

Isaac Mubiru

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PROFESSIONAL SUMMARY

Full-stack engineer with 6+ years of fully remote work who owns problems end to end — **React interfaces, Node.js backend services and APIs, asynchronous processing on cloud infrastructure**, and running the result in production. Led the zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, reporting to the CTO, while the live product kept running. Shipped a customer-facing **video feature** (ffmpeg overlay) that increased buyer conversion, a production **LLM assistant** (OpenAI + Google Vision OCR over WhatsApp), and an **agentic pipeline** on Claude Code that verifies inputs before any model call. Now applying that ownership to agent-powered creative tools. Code: github.com/zac-09.

CORE COMPETENCIES

End-to-End Feature Ownership

React Frontend Delivery

Node.js Backend Services & APIs

Asynchronous Processing (Pub/Sub, Kafka)

Cloud Infrastructure (GCP, Firebase)

LLM Integration & Agentic Workflows

Video / Media Processing (ffmpeg)

Zero-Downtime Production Operation

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote · e-commerce

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Built a customer-facing **3D visualisation feature** using video overlay and the **ffmpeg** library, increasing buyer conversion.
- Implemented real-time two-way sync between MongoDB and Firestore using **Node.js and Google Pub/Sub** for asynchronous message processing.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a Python script; saved **\$5,000/month** by moving all services off AWS.
- Shipped Express Shipping (+\$40 revenue per order) and migrated a Webflow website from a Parse backend to Firebase.

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Built the migration script that read data from MongoDB, ran complex processing logic and wrote to Firestore.
- Wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python); trained the Ops team on the Firebase dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote · fintech PWA

- Led optimisation of a PWA delivering real-time stock prices; built ad-hoc jobs over **2M+ Firestore records** to keep prices current from Morningstar APIs.
- Worked across the stack: responsive UI pages in **React** and Tailwind, and backend APIs in **TypeScript** on Firebase Cloud Functions and Express.js.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote · agency

- Full-stack delivery for demanding clients, reporting to the CTO: **React UIs from Figma designs** under tight timelines and Node.js backends for DeFi applications using Web3.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote · legal tech

- Led a team of 4 developers building complete production systems for NGOs: a case management system (FIDA Uganda) and a paralegal database (LASPNET Uganda).
- Architected a microservices backend on **Apache Kafka, Docker and Kubernetes**; built a USSD service talking to a Node.js backend over gRPC.
- Developed a cross-platform mobile app in React Native and Expo (iOS & Android) with Firebase push notifications.

PROJECTS

Agentic Evaluation Pipeline agents

Extended an open-source agentic pipeline on Claude Code that scans job portals, verifies each source live in a headless browser before any model scoring, deduplicates inputs and generates tailored documents — 200+ evaluations run with verification-before-inference as the core rule.
Claude Code · Playwright · Node.js · YAML

Healthcare WhatsApp Assistant multimodal LLM

Conversational assistant that reads prescription photos and replies inside WhatsApp: OpenAI for dialogue, Google Vision for OCR, behind the WhatsApp Business API.
OpenAI API · Google Cloud Vision · WhatsApp Business API · Node.js

Real-time Pub/Sub Sync Pipeline

Live MongoDB↔Firestore sync supporting parallel production traffic during a zero-downtime, 20+ application migration.
Node.js · Google Pub/Sub · MongoDB · Firestore

EDUCATION

B.Sc. Computer Engineering — Makerere University 2018 – 2022
Kampala, Uganda

SKILLS

Languages: JavaScript, TypeScript, Python, HTML, CSS, Go (intermediate), Rust (intermediate)
Frontend: React, Tailwind, PWA, React Native (intermediate)
Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Firebase Cloud Functions
AI: OpenAI API, Google Cloud Vision, Claude Code agentic workflows
Async & Data: Google Pub/Sub, Apache Kafka, NATS Streaming, MongoDB, Firestore
Cloud / Infra: GCP, Firebase, Google Cloud Storage, AWS (intermediate), Docker, Kubernetes, Linux, Git **Media:** ffmpeg (video overlay)
Spoken: English, Kiswahili